

## Lab 1: DevOps Foundations & Continuous Integration

1. Record all commands

Record all commands, pipeline configurations, screenshots, commit history, and observations for each experiment.

### Docker Commands

1. Check Docker Installation: `docker --version`

A terminal window with a dark background. The title bar shows 'ayushi2323@LAPTOP-6LVIDNI'. The command 'docker --version' has been executed, and the output is 'Docker version 29.7.2, build a7dcaa6'.

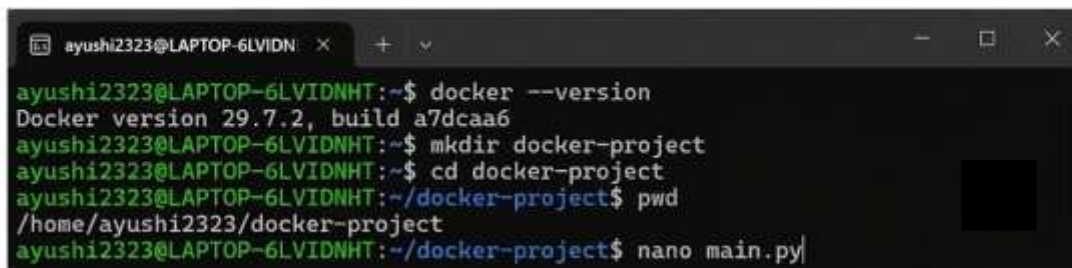
```
ayushi2323@LAPTOP-6LVIDNHT:~$ docker --version
Docker version 29.7.2, build a7dcaa6
```

2. Create Project Directory: `mkdir docker-project`, `cd docker-project`, `pwd`

A terminal window showing the execution of three commands: 'mkdir docker-project', 'cd docker-project', and 'pwd'. The output of 'pwd' is '/home/ayushi2323/docker-project'.

```
ayushi2323@LAPTOP-6LVIDNHT:~$ mkdir docker-project
ayushi2323@LAPTOP-6LVIDNHT:~$ cd docker-project
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ pwd
/home/ayushi2323/docker-project
```

3. Create Python Application: `nano main.py`

A terminal window showing the full sequence of commands from the previous steps. The last command entered is 'nano main.py'.

```
ayushi2323@LAPTOP-6LVIDNHT:~$ docker --version
Docker version 29.7.2, build a7dcaa6
ayushi2323@LAPTOP-6LVIDNHT:~$ mkdir docker-project
ayushi2323@LAPTOP-6LVIDNHT:~$ cd docker-project
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ pwd
/home/ayushi2323/docker-project
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ nano main.py
```

4. List Docker Images: docker images

```
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ docker images
```

| IMAGE                         | ID           | DISK USAGE | CONTENT SIZE | Info → | U | in Use |
|-------------------------------|--------------|------------|--------------|--------|---|--------|
| docker-demo:1.0               | 3b67be2e461d | 224MB      | 55.1MB       |        | U |        |
| hello-world:latest            | 5dd0d3e6e255 | 25.9kB     | 9.49kB       |        |   |        |
| my-docker-app:latest          | ch9044724ddb | 1.62GB     | 418MB        |        |   |        |
| nginx:latest                  | 05b8cb60c354 | 253MB      | 69.2MB       |        | U |        |
| python:3.12-slim              | 0977da3bc194 | 179MB      | 45.4MB       |        |   |        |
| redis:latest                  | 298e5b3bc566 | 212MB      | 57.5MB       |        |   |        |
| srsihti2323/my-nontain::x:1.0 | 05b8cb60c354 | 253MB      | 69.2MB       |        | U |        |
| ubuntu:latest                 | 2260313b31c8 | 160MB      | 45.3MB       |        | U |        |

5. Run Docker Containers: docker run -d -p 5000:5000 - name my container mydocker-app

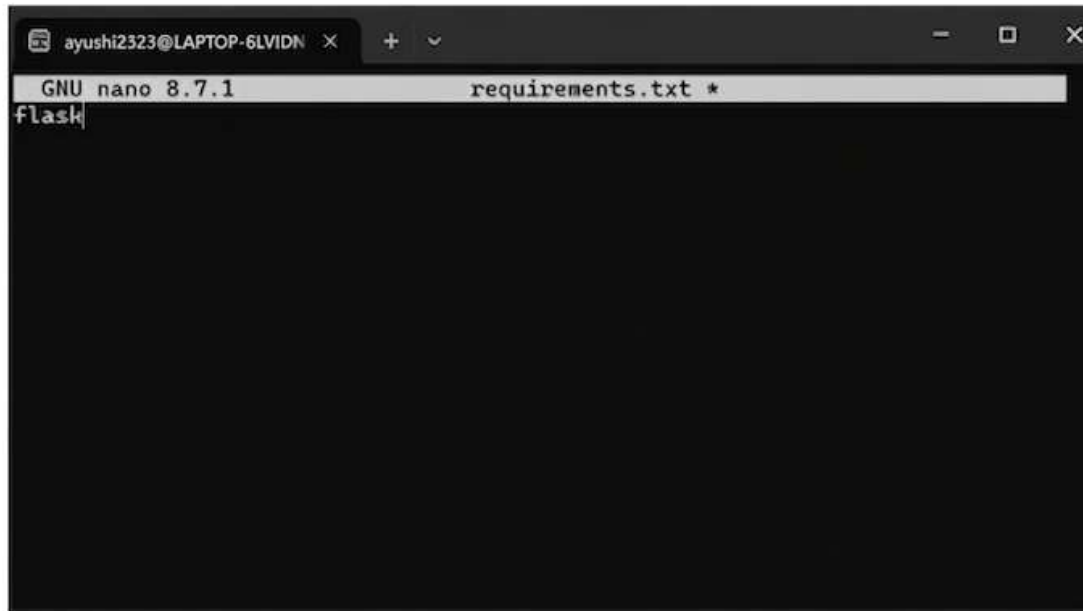
```
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ docker run -d -p 5000:5000 --name my-container my-docker-app
8f3b05f4599b0bb875651cd4fb70f52da9c71a5ef5c2321b3b9c50d080bfd0459
```

7. Stop the Docker Container: docker stop my-container

```
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ docker stop my-container
my-container
```

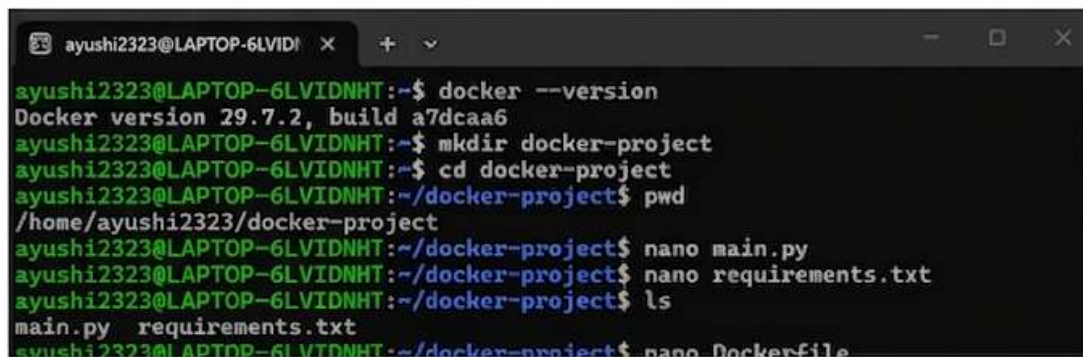
8. Remove the Docker Image: docker rmi ayushi2323/my-docker-app

```
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ docker rmi ayushi2323/my-docker-app
Untagged: ayushi2323/my-docker-app:latest
Untagged: ayushi2323/my-docker-app:latest
Deleted: sha256:d6d02a927d6d1d4d6a5f780517f8a7d2e0b5c17d7b0f0a4f5f5f5f5f5f5f5f5f
... (more 'Deleted' lines)
... (more sha256:d6d02a927d6d1d4d6a5f780517f8a7d2e0b5c17d7b0f0a4f5f5f5f5f5f5f5f5f
... (more sha256:d6d02a927d6d1d4d6a5f780517f8a7d2e0b5c17d7b0f0a4f5f5f5f5f5f5f5f5f
```



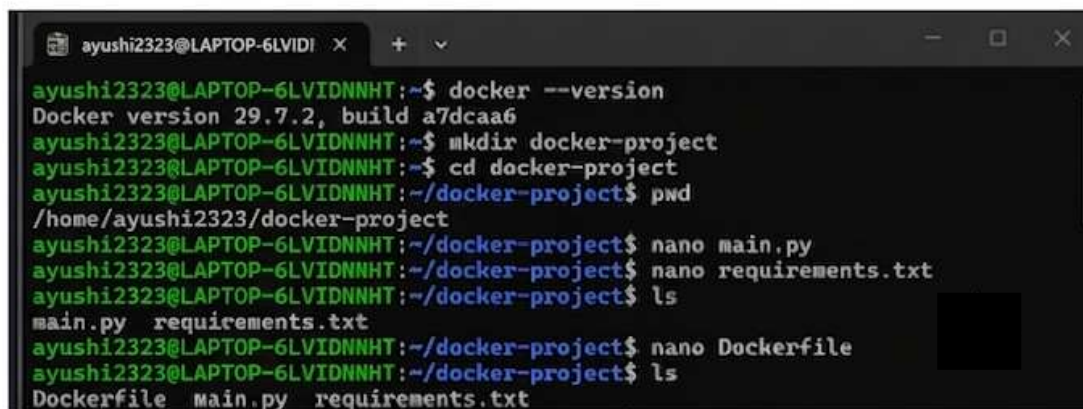
```
ayushi2323@LAPTOP-6LVIDN x + v
GNU nano 8.7.1 requirements.txt *
flask
```

2. Create Dockerfile: `nano Dockerfile`



```
ayushi2323@LAPTOP-6LVIDNHT:~$ docker --version
Docker version 29.7.2, build a7dcaa6
ayushi2323@LAPTOP-6LVIDNHT:~$ mkdir docker-project
ayushi2323@LAPTOP-6LVIDNHT:~$ cd docker-project
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ pwd
/home/ayushi2323/docker-project
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ nano main.py
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ nano requirements.txt
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ ls
main.py requirements.txt
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ nano Dockerfile
```

3. List Project Files: `ls`



```
ayushi2323@LAPTOP-6LVIDNHT:~$ docker --version
Docker version 29.7.2, build a7dcaa6
ayushi2323@LAPTOP-6LVIDNHT:~$ mkdir docker-project
ayushi2323@LAPTOP-6LVIDNHT:~$ cd docker-project
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ pwd
/home/ayushi2323/docker-project
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ nano main.py
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ nano requirements.txt
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ ls
main.py requirements.txt
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ nano Dockerfile
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ ls
Dockerfile main.py requirements.txt
```

4. Build Docker Image: `docker build -t my-docker-app.`

4. List Docker Images: docker images

```
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ docker images
```

| IMAGE                         | ID           | DISK USAGE | Info →       | U      | in Use |
|-------------------------------|--------------|------------|--------------|--------|--------|
| docker-demo:1.0               | 3b67be2e461d | 224MB      | CONTENT SIZE | 55.1MB | U      |
| hello-world:latest            | 5dd0d3e6e255 | 25.9kB     | 9.49kB       |        | U      |
| my-docker-app:latest          | ch9044724ddb | 1.62GB     | 418MB        |        |        |
| nginx:latest                  | 05b8cb60c354 | 253MB      | 69.2MB       | U      |        |
| python:3.12-slim              | 0977da3bc194 | 179MB      | 45.4MB       |        |        |
| redis:latest                  | 298e5b3bc566 | 212MB      | 57.5MB       |        |        |
| srsihti2323/my-nontain::x:1.0 | 05b8cb60c354 | 253MB      | 69.2MB       | U      |        |
| ubuntu:latest                 | 2260313b31c8 | 160MB      | 45.3MB       | U      |        |

5. Run Docker Containers: docker run -d -p 5000:5000 - name my container mydocker-app

```
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ docker run -d -p 5000:5000 --name my-container my-docker-app
8f3b05f4599b0bb875651cd4fb70f52da9c71a5ef5c2321b3b9c50d080bfd0459
```

7. Stop the Docker Container: docker stop my-container

```
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ docker stop my-container
my-container
```

8. Remove the Docker Image: docker rmi ayushi2323/my-docker-app

```
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ docker rmi ayushi2323/my-docker-app
Untagged: ayushi2323/my-docker-app:latest
Untagged: ayushi2323/my-docker-app:latest
Deleted: sha256:d6d02a927d6d1d4d6a5f780517f8a7d2e0b5c17d7b0f0a4f5f5f5f5f5f5f5f5f
... (more 'Deleted' lines)
... (more sha256:d6d02a927d6d1d4d6a5f780517f8a7d2e0b5c17d7b0f0a4f5f5f5f5f5f5f5f5f
... (more sha256:d6d02a927d6d1d4d6a5f780517f8a7d2e0b5c17d7b0f0a4f5f5f5f5f5f5f5f5f
```

```
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ git commit -m "Initial Docker project"
[master (root-commit) 9ec3c01] Initial Docker project
5 files changed, 67 insertions(+)
create mode 100644 Dockerfile
create mode 100644 commands.txt
create mode 100644 main.py
create mode 100644 observations.txt
create mode 100644 requirements.txt
```

18. Update Observations File: [nano observations.txt](#)

```
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ nano observations.txt
```

```
GNU nano 8.7.1 observations.txt *
Experiment: Containerization of a Sample Application

Objective:
To containerize a sample Python application using Docker.

Observations:

1. Docker was successfully installed and verified.
2. A Python application was created.
3. A Dockerfile was created to define the application environment.
4. A Docker image was successfully built.
5. A container was created from the Docker image.
6. Port 5000 was mapped from the container to the host.
7. The application was successfully accessed through the browser.
8. Docker history displayed the image build layers.
9. The container was stopped and started successfully.
10. Container logs were viewed using docker logs.

Result:
The sample Python application was successfully containerized
and executed using Docker.
```

19. Add Updated Files: [git add .](#)

```
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ nano observations.txt
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ git add .
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ git commit -m
git commit -m "Updated Docker experiment observations"
On branch master
nothing to commit, working tree clean
```

20. Create Second Commit: `git commit -m "Updated Docker experiment observations"`

```
ayushi2323@LAPTOP-6LVIDNNHT:~/docker-project$ git commit -m "Initial Docker project"
[master (root-commit) 9ec3c01] Initial Docker project
 5 files changed, 67 insertions(+)
 create mode 100644 Dockerfile
 create mode 100644 commands.txt
 create mode 100644 main.py
 create mode 100644 observations.txt
 create mode 100644 requirements.txt
```

18. View Commit History: `git log --oneline`, `git status`

```
ayushi2323@LAPTOP-6LVIDNNHT:~/docker-project$ git log --oneline
9ec3c01 (HEAD -> master) Initial Docker project
ayushi2323@LAPTOP-6LVIDNNHT:~/docker-project$ git status
On branch master
nothing to commit, working tree clean
ayushi2323@LAPTOP-6LVIDNNHT:~/docker-project$
```